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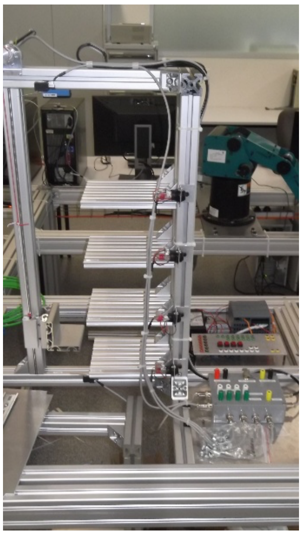
LAB REPORT

Subject: MECH470/970

Warehouse Lift System



Week 7

Group 1
Team A

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Objective

The objective of this lab is to design and program a ladder logic diagram for a warehouse lift system. The equipment included a DC motor, a laser distance sensor, 4 indicators, and 4 push-button switches on the elevator. To complete this, a control strategy was chosen to handle input requests and direct the lift to the corresponding floor.

Lab Tasks and Design Justification

Determine the Lift Position using the laser sensor

Based on the sensor datasheet and lab information, the laser distance sensor outputs a 4-20mA signal proportional to the distance, which is then passed through a 500 Ω resistor to produce a voltage that can be read on the analogue input of the PLC for a value between 0 and 32767. To convert this analogue signal to the distance in mm, the lift was moved to the very top and bottom, and the analogue value was determined at each extremity experimentally. Then a 'normalise' block was used to change the bottom of the lift to the value 0 and the top to 100. This was then fed into a scale block that scales the 0 to 100 to a value of 0 to 677 mm. The value of the output-scaled position was saved to a memory location 'MD100' so it could be accessed by the subsequent ladder rungs to determine the position of the lift.

Create a control program to move the lift to a certain position using the DC motor

The motor was connected to the analogue output 'QW80' that sends the signal to the motor. Through reviewing the motor information and experimentation, it was determined that sending a signal of ~16,000 would stop the motor, a signal of ~32,000 would wind the motor and lift upwards, and 0 would wind the motor and lift downwards.

The control strategy used was two-position control. A neutral zone of approximately 20 mm was created to avoid the laser and motor hysteresis. The logic used to control the lift and motor was, if blocks and move blocks. For example, if the laser is telling the PLC that the lift is below the desired location, the motor will be sent 32,000, which will raise the lift. Once it is in the 20 mm neutral zone around the desired location, the motor is sent a signal of 16,000, which stops the motor and the lift.

Read in the pushbutton switches to actuate the lift to the level of the pushbutton activated

To achieve reading the pushbutton switches and controlling the lift to each level, first, the height of each level was experimentally determined. Once the 4 values for each floor height were found, there were 4 ladder rungs that were each in control of moving the lift to one of the levels.

These rungs control the motor to the correct floor by using greater than and less than blocks. For example, floor 2 is around the height of 400-420 mm. If the rung to send the lift to that floor is activated, the greater than and less than blocks will look at the current position of the lift from the laser sensor, if it is less than 400, the motor is turned on to bring the lift up, and then once the laser indicates the lift is within the neutral zone, it is switched off.

The pushbuttons were handled by using NO switches that latch when activated, which also turns on the indicator for that floor. There is a 'NC reset switch' for each rung that is connected to a reset Boolean temporary value. After a pushbutton is activated, it also activated the ladder rung below that is controlling the motor to that level. And when the motor has successfully been moved to within the neutral zone of the correct level, the motor is switched off (given a signal of 16,000) and the reset temporary value is set to true, which then unlatches the indicator, thus turning off the indicator and powering off the rung that moves the motor to that floor.

An additional ladder rung was implemented that continuously sent a signal of 16,000 (no movement) to the DC motor if no pushbuttons were activated, so the motor remained stationary if it was not required to move. This control strategy successfully moved the lift to the correct floor when the pushbutton was pressed, thus completing the required tasks.

Extra Activities

- *Create a system to handle multiple lift requests (completed)*

The control strategy used and described above already had the capability to handle multiple lift requests with a built-in priority. Every time a pushbutton was pressed, it latched, thereby allowing more than one indicator to be activated at a time. Then the lift was moved to each required floor location and unlatched the floors as it went there. The priority was given to the upper levels. As level four was the first in the ladder rungs, the motor would move there first even if a lower level was pressed before it.

- *Implement a two-stage system to indicate that the lift request has been registered (blinking) and is enroute (stable) (not implemented)*

A two-stage blinking and enroute indicator system was not implemented within the allotted lab time. Due to the design of the control system, it was difficult to determine which floor the lift was travelling to.

In the future, to complete this task, a memory location of 8 bits could be used as a queue and enroute register. So if the pushbutton was activated but the lift was already moving somewhere else, this request could be sent to the queue, where the indicator light would be flashing while it is in the queue list and stable when it is moved from the queue list to the enroute list.

- *Develop a self-diagnosis system to alert if the laser has failed or the elevator actuator is not functioning (completed)*

A self-diagnosis system was implemented with a less-than block that would activate one of the hard-wired output indicators. If the laser sensor outputs a value that was outside the determined correct range, an error output indicator was activated.

Reflection & Conclusion

Design challenges and lessons learned

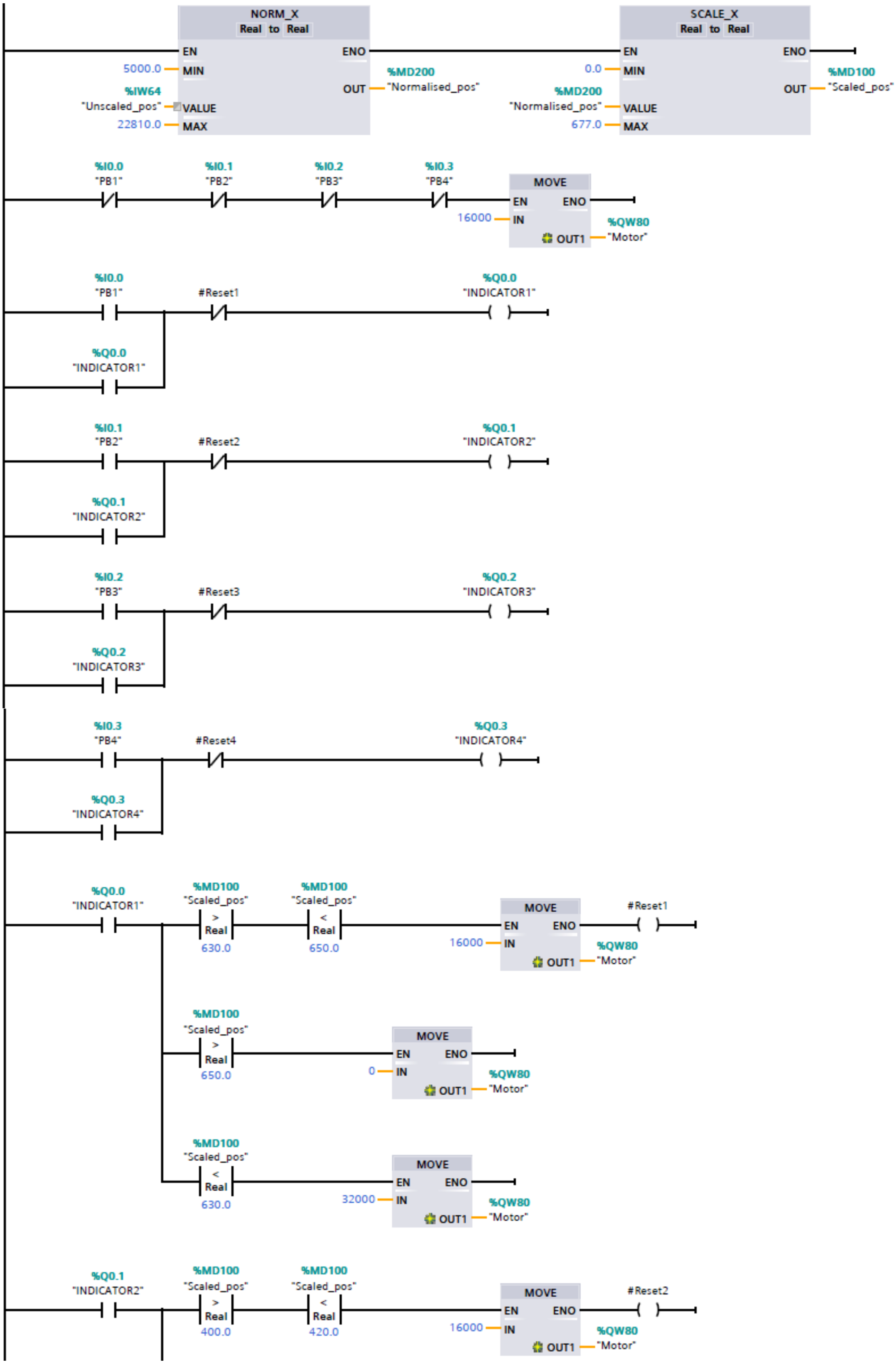
All the main tasks and 2 out of the 3 extra tasks were completed successfully and demonstrated to the lab demonstrators within the allotted lab time. The ladder rung diagram that was constructed was robust, and the lift performed very well during experimentation. An initial error was not using a neutral zone, as this caused the motor to quickly turn on and off as it tried to get the exact location of the floor.

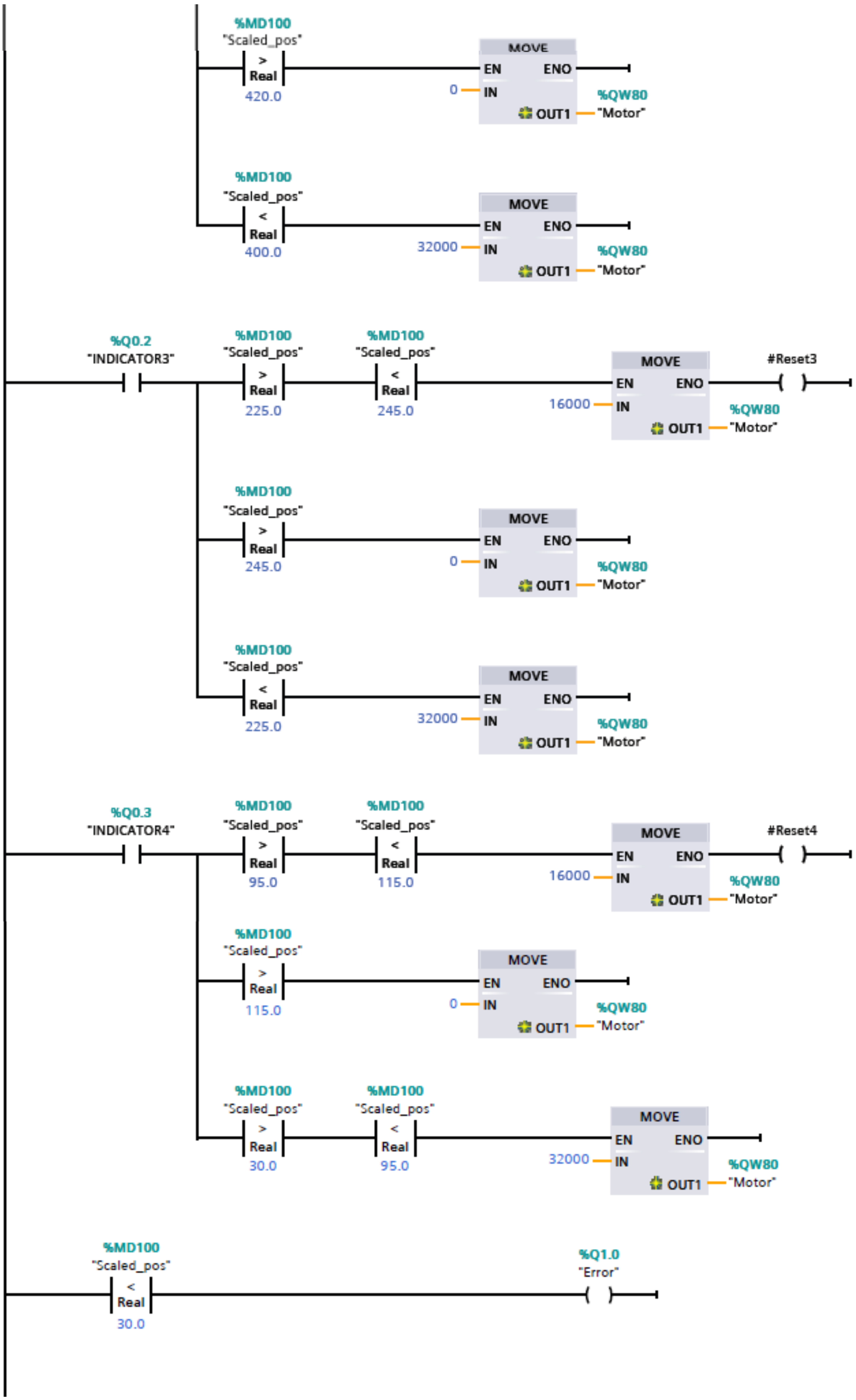
Another challenge was that the strategy used did not allow for easy expansion. Specifically for the second extra task of a two-stage blinking indicator system. In the future, using the available memory to robustly keep track of where the lift is moving to and completing that move before getting the next request from the queue would be more realistic to an actual lift program.

In the future, additional planning will be performed to ensure the initial strategy can be expanded easily to complete even the extra tasks.

Appendix

A. Ladder Logic Diagram




%MD100
"Scaled_pos"
>
Real
420.0
MOVE
EN
ENO
0
IN
%QW80
OUT1
"Motor"
%MD100
"Scaled_pos"
<
Real
400.0
MOVE
EN
ENO
32000
IN
%QW80
OUT1
"Motor"
%Q0.2
"INDICATOR3"
%MD100
"Scaled_pos"
>
Real
225.0
%MD100
"Scaled_pos"
<
Real
245.0
MOVE
EN
ENO
#Reset3
16000
IN
%QW80
OUT1
"Motor"
%MD100
"Scaled_pos"
>
Real
245.0
MOVE
EN
ENO
0
IN
%QW80
OUT1
"Motor"
%MD100
"Scaled_pos"
<
Real
225.0
MOVE
EN
ENO
32000
IN
%QW80
OUT1
"Motor"
%Q0.3
"INDICATOR4"
%MD100
"Scaled_pos"
>
Real
95.0
%MD100
"Scaled_pos"
<
Real
115.0
MOVE
EN
ENO
#Reset4
16000
IN
%QW80
OUT1
"Motor"
%MD100
"Scaled_pos"
>
Real
115.0
MOVE
EN
ENO
0
IN
%QW80
OUT1
"Motor"
%MD100
"Scaled_pos"
>
Real
30.0
%MD100
"Scaled_pos"
<
Real
95.0
MOVE
EN
ENO
32000
IN
%QW80
OUT1
"Motor"
%MD100
"Scaled_pos"
<
Real
30.0
%Q1.0
"Error"